

P74 Series Differential Pressure Controls

Application

These differential pressure controls are for use as operating controls and/or indicating system functions through display lights or panels. They measure the difference in pressure exerted upon its two sensing elements.

The controls are available for applications sensing air, oil or liquid. Typical applications are proof of flow across a chiller or water cooled condenser, proof of flow in a heating or cooling coil and lube oil pressure sensing on refrigeration compressors. On a proof of flow application the control measures pressure drop across two different points in either a closed water circulating system or a city water to supply system.



Fig. 1: P74 Differential Pressure Control with Style 13 elements.

Specifications

Type Number	P74AA	SPST, Contacts Open on Differential Pressure Increase
	P74AB	SPST, Contacts Open on Differential Pressure Increase, Manual Reset
	P74BA	SPST, Contacts Close on Differential Pressure Increase
	P74CA	DPST, Contacts Open on Differential Pressure Increase
	P74CB	DPST, Contacts Open on Differential Pressure Increase, Manual Reset
	P74DA	DPST, Contacts Close on Differential Pressure Increase
	P74DB	DPST, Contacts Close on Differential Pressure Increase, Manual Reset
	P74EA	SPDT, Standard Switch Differential
	P74FA	SPDT, Narrow Switch Differential
	P74GA	Main Contacts Open on Differential Increase, Separate Auxiliary Contacts Close
	P74HA	Main Contacts Close on Differential Increase, Separate Auxiliary Contacts Open
Ambient Temperature	Minimum	32°F (0°C)
	Maximum	104°F (40°C)
Conduit Opening	7/8" (22 mm) Diameter Hole for 1/2" Conduit	
Contact Unit	P74A, P74B, P74C, P74D, P74G, P74H	Large Copper Backed Silver Contacts and Beryllium Copper Conductor Leaves
	P74E, P74F	SPDT, Snap-Acting Contacts in Dust Protected Enclosure
Differential	See Table	
Finish	Gray Baked	
Material	Case	.062" (1.6 mm) Cold Rolled Steel
	Cover	.028" (0.7 mm) Cold Rolled Steel
Maximum Pressure	See Table	
Mounting Bracket	Universal Mounting Bracket Part No. 271-51 Supplied as Standard	
Range	See Table	
Shipping Weight	Individual Pack	2.4 lb (1.1 kg)
	Overpack of 20	50.5 lb (22.9 kg)
	Bulk Pack of 25	57 lb (25.9 kg)

On a proof of flow application in a water chiller system the control activates a light or signal to indicate a loss of water.

The control may also be applied as a lube oil pressure sensing control on refrigeration compressors. They may be used in combination with P28 and/or P45 oil pressure cutout controls on two compressor, single motor units to reduce the oil system cost. (See Fig. 4.) Special low pressure models are available for variable speed and screw compressor oil pressure applications.

All Series P74 differential pressure controls are designed for use *only* as operating controls. Where an operating control failure would result in personal injury and/or loss of property, it is the responsibility of the installer to add devices (safety, limit controls) or systems (alarm, supervisory systems) that protect against, or warn of, control failure.

Range and Differential Specifications

Differential Pressure Range	Switch Differential			Maximum Differential Pressure Between the Bellows	Maximum Low Pressure Bellows Overrun*	Bellows Material
	<u>PSI</u> kPa	<u>PSI</u> kPa				
	P74A, P74B, P74C, P74D, P74G, P74H	P74E	P74F	<u>PSI</u> kPa	<u>PSIG</u> kPa	
<u>2 to 26</u>	—	<u>3.5 Fixed</u>	<u>1.2 Fixed</u>	120	180	Brass
14 to 180		24	8	830	1241	
<u>8 to 60</u>	6 to 20 Adj.	<u>3.8 Fixed</u>	<u>1.5 Fixed</u>	120	180	Brass
50 to 400	41 to 138	26	10	830	1241	
<u>2 to 30</u>	—	<u>5.0 Fixed</u>	<u>2.0 Fixed</u>	200	180	Stainless Steel
14 to 207		34	14	1379	1241	
<u>8 to 70</u>	8 to 30 Adj.	<u>5.5 Fixed</u>	<u>2.5 Fixed</u>	200	180	Stainless Steel
50 to 450	55 to 207	38	17	1379	1241	

*Bellows overrun pressure is the pressure supplied to the low pressure side of the control.

Features

- Heavy duty, low profile elements withstand unduly high overrun pressures that may be encountered in shipment or in some machine rooms.
- Lockout models have a “trip-free” manual reset.
- Long life contact structure with high contact force -- no contact bounce.
- Single unit mounting and wiring -- saves installation time and material.

General Description

Single and double pole models are available with contacts that open on a pressure differential increase or close on a pressure differential increase. Also available are models with single-pole, double-throw enclosed contacts or with main and separate reverse-acting auxiliary contacts. Controls with lockout feature require manual reset to reclose circuit after lockout. The “trip-free” reset will not permit restart until reset button is pushed and released.

The operation point of the control is readily adjusted by rotating the adjusting disk. The control set points are easily read on a calibrated scale.

Electrical Ratings

P74AA, P74AB, P74BA

Motor Ratings	1 Phase			
	120 V	208 V	240 V	277 V
AC Full Load Amp	20.0	18.7	17.0	—
AC Locked Rotor Amp	120.0	112.2	102.0	—
AC Non-Inductive Amp	22.0	22.0	22.0	—
Pilot Duty — 125 VA, 120 to 600 VAC; 57.5 VA, 120 to 300 VDC				

P74CA, P74CB, P74DA, P74DB

Motor Ratings	1 Phase				Polyphase	
	120 V	208 V	240 V	277 V	208 V	240 V
Horsepower	2	3	3	—	5	5
AC Full Load Amp	24.0	18.7	17.0	—	16.5	15.0
AC Locked Rotor Amp	144.0	112.2	102.0	—	99.0	90.0
AC Non-Inductive Amp	24.0	24.0	24.0	22.0	—	—
Pilot Duty — 125 VA, 120 to 600 VAC; 57.5 VA, 120 to 300 VDC						

P74EA

Motor Ratings	120 V	208 V	240 V	277 V
AC Full Load Amp	16.0	9.2	8.0	—
AC Locked Rotor Amp	96.0	55.2	48.0	—
AC Non-Inductive Amp	16.0	16.0	16.0	16.0
Pilot Duty — 125 VA, 120 to 600 VAC				

P74FA

Motor Ratings	120 V	208 V	240 V	277 V
AC Full Load Amp	6.0	3.4	3.0	—
AC Locked Rotor Amp	36.0	20.4	18.0	—
AC Non-Inductive Amp	10.0	10.0	10.0	10.0
Pilot Duty — 125 VA, 120 to 277 VAC				

P74GA, P74HA

Pole Number	LINE-M2 (Main)				LINE-M1 (Auxiliary)			
	120 V	208 V	240 V	277 V	120 V	208 V	240 V	277 V
AC Full Load Amp	16.0	9.2	8.0	—	6.0	3.3	3.0	—
AC Locked Rotor Amp	96.0	55.2	48.0	—	36.0	19.8	18.0	—
AC Non-Inductive Amp	16.0	9.2	8.0	7.2	6.0	6.0	6.0	6.0
Pilot Duty, Both Poles — 125 VA, 120 to 600 VAC; 57.5 VA, 120 to 300 VDC								

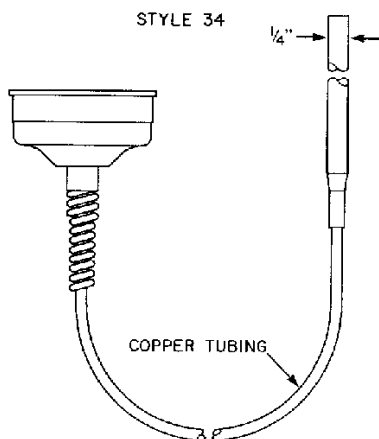
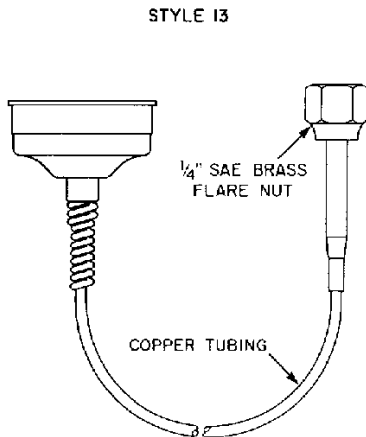
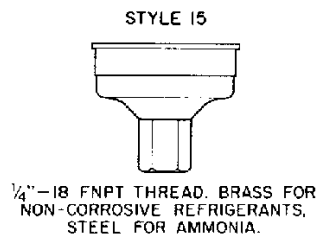
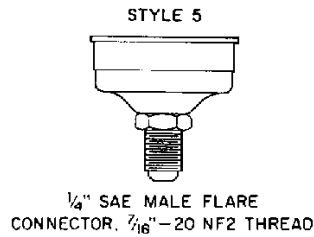


Fig. 2: Pressure element styles available on the P74. Style 13 is standard. Other styles shown above can be supplied on quantity orders.

Optional Constructions

Pressure Elements

Regularly supplied for non-corrosive refrigerants (fluorinated hydrocarbons). Available for ammonia service with 1/4 in. - 18 FNPT connector (See Style Chart, Fig. 2.)

Pressure Connectors

Standard controls supplied with 36 in. capillary tubing with 1/4 in. flare nut (Style 13). Controls with 1/4 in. SAE male flare connector (no capillary tubing, Style 5), 36 in. capillary tubing with 1/4 in. sweat section (Style 34), or 1/4 in. FNPT connector (Style 15) may be supplied on quantity orders (see Pressure Element Styles).

Repairs and Replacement

Field repairs must not be made. For a replacement control, contact the nearest Johnson Controls distributor.

Ordering Information

To order, specify:

1. Quantity required.

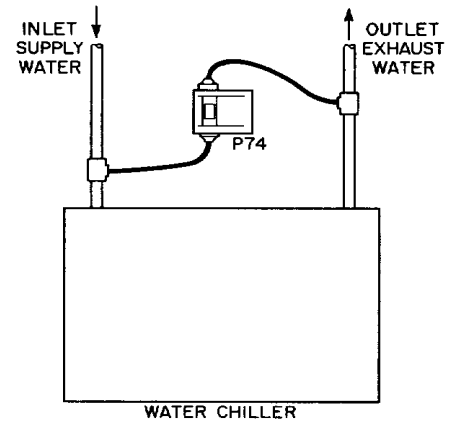


Fig. 3: Typical proof of flow hookup.

2. Complete Product Number, if available.
3. If complete Product Number is not available, specify Type Number (see Specifications table) and the following.
4. Type of refrigerant or fluid.
 - a. Non-corrosive.
 - b. Ammonia.
5. Style of pressure connector.
6. Optional constructions.
7. Setting -- contacts close at ____ and open at ____.

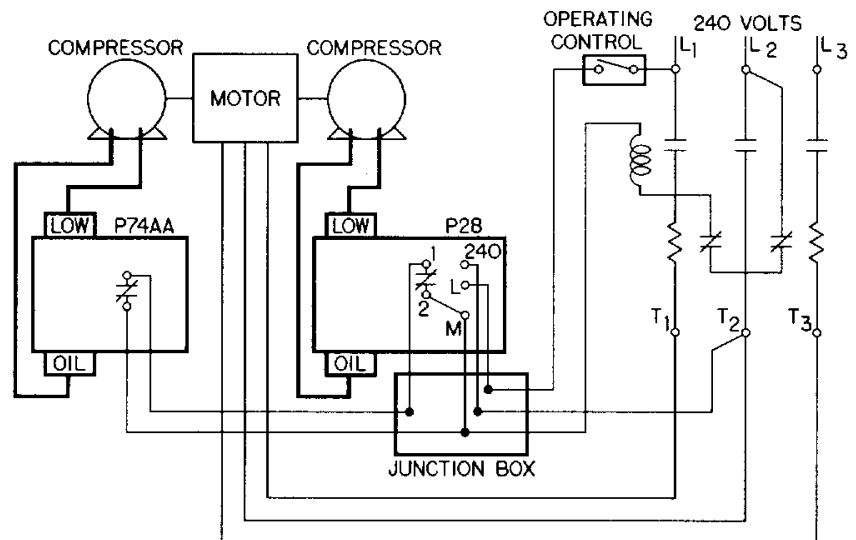
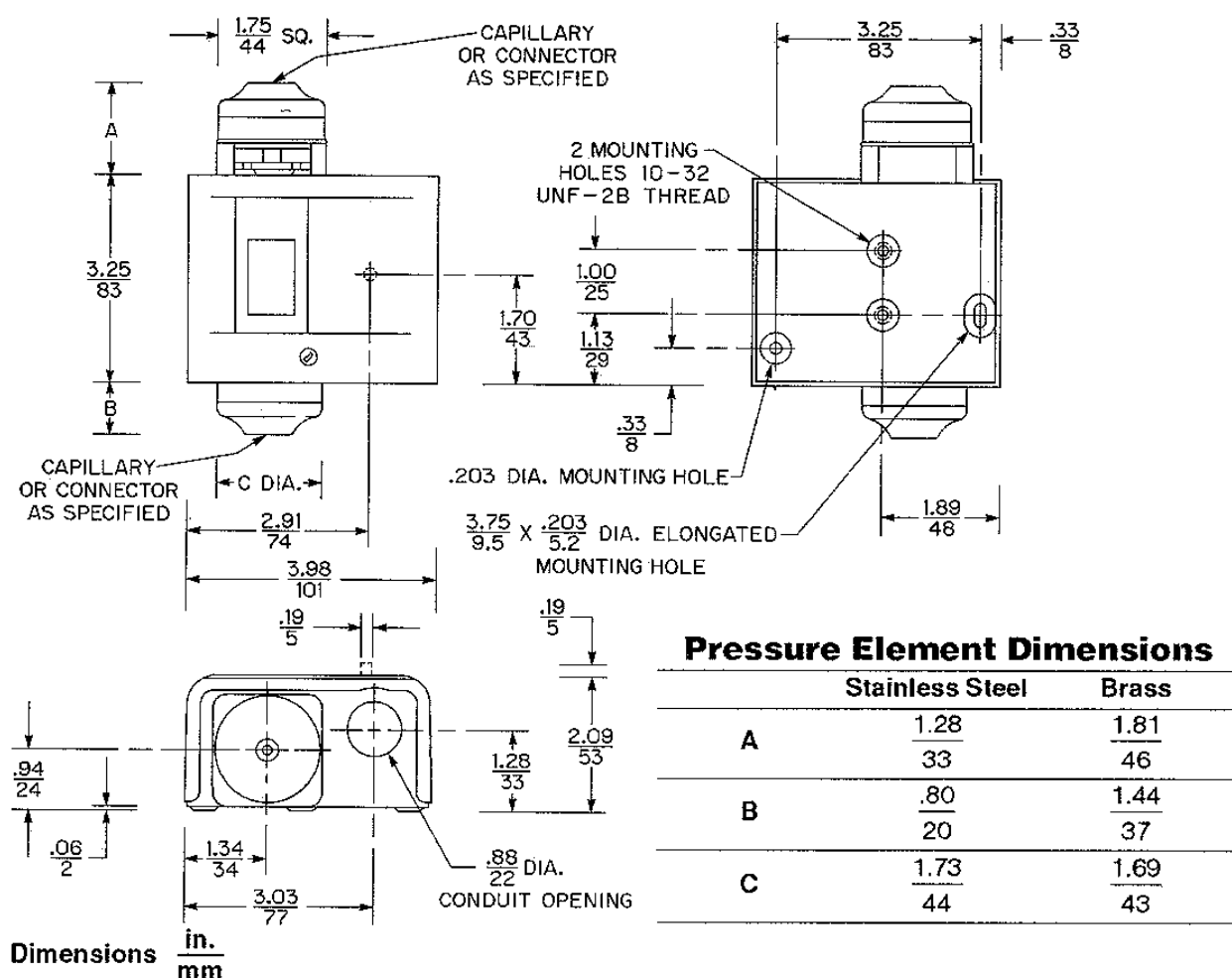


Fig. 4: Typical wiring diagram showing the P74AA and a P28 on a motor operating two compressors.



Performance specifications appearing herein are nominal and are subject to accepted manufacturing tolerances and application variables.

European Single Point of Contact:

JOHNSON CONTROLS
WESTENDHOF 3
45143 ESSEN
GERMANY

NA/SA Single Point of Contact:

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507 E MICHIGAN ST
MILWAUKEE WI 53202
USA

APAC Single Point of Contact:

JOHNSON CONTROLS
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NO. 22 BLOCK D NEW DISTRICT
WUXI JIANGSU PROVINCE 214142
CHINA



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